Nakshatra Aripineni

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Education

Bachelor of Science in Computer Science | *The University of Texas at Dallas* Expected May 2029
Relevant Coursework: Data Structures & Algorithms, Computer Science I & II, Discrete Mathematics, Linear Algebra, Object-Oriented Programming, Foundations of Computer Science, Calculus I

Technical Skills

Programming Languages: Python, Java, JavaScript, C++, C, HTML/CSS
Frameworks & Libraries: React.js, Node.js, PyTorch, scikit-learn, Pandas, NumPy
Databases & Cloud: SQL, PostgreSQL, AWS, Docker, Cloudflare
AI & Developer Tools: Claude Code, Cursor, GitHub Copilot, Git/GitHub, Linux/Unix, IBM Bob

Technical & Research Experience

Software Engineer | *IBM SkillsBuild AI Experiential Learning Lab* Oct 2026 – Present
- Architecting **Finality**, a multi-agent trust layer that autonomously approves, escalates, or blocks AI-initiated payments.
- Training fraud-detection and cash-forecasting models that power **4** agents evaluating invoice, liquidity, and policy risk.
- Operationalizing human-in-the-loop manager approvals and an explainable audit timeline that traces every payment decision.

AI/ML Developer | *IEEE* Oct 2026 – Present
- Constructing the **AI-Powered Skill Gap Analyzer**, a full-stack app that benchmarks student skills against job postings.
- Applying **natural language processing** to parse job descriptions, extract required skills, and rank gaps by career impact.
- Delivering a user-centered interface with API integrations that convert skill gaps into a prioritized learning roadmap.

Full Stack Developer | *Nebula Labs* Sep 2026 – Present
- Developing **UTD Clubs**, an open-source platform unifying **400+** student organizations into one searchable directory.
- Designing responsive interfaces with dynamic filtering, an events calendar, and an AI-powered club-matching questionnaire.
- Implementing multi-provider **OAuth** (Google, Discord) and a Nebula API caching layer for secure, low-latency data access.

Machine Learning Engineer | *Google Developer Group* Sep 2026 – Present
- Directing Python and ML development for **Green Light**, a real-time adaptive traffic-signal system that reduces congestion.
- Modeling traffic dynamics with **PyTorch** to forecast turning movements, queues, and pedestrian flow **15–60 seconds** ahead.
- Orchestrating a **C++** engine and React dashboard that vet predictions against traffic rules before adjusting signals.

Undergraduate Student Researcher | *RIDE Program, UT Dallas* Aug 2026 – Present
- Investigating **LLM**-driven program repair, code generation, and software testing to advance intelligent developer tooling.
- Prototyping AI-assisted refactoring workflows that improve code quality, maintainability, and developer productivity.
- Evaluating how language models comprehend large codebases to enable context-aware debugging and repository understanding.

Projects

Synapse | *Code Architecture Visualizer* Aug 2026
- Created a developer tool that parses code in **5 languages** (C++, TypeScript, JavaScript, Python, Go) into visual diagrams.
- Mapped function and component relationships into interactive graphs with multiple layout options and automated testing.

CodeGuard AI | *Code Quality & Vulnerability Analyzer* Jul 2026
- Built a **Gemini API** analyzer that detects syntax errors, security vulnerabilities, and quality issues across **3 languages**.
- Crafted structured prompts that supply code context to an LLM and generate precise, actionable refactoring recommendations.

Leadership & Involvement

Mentoring Team Manager | *Women Mentoring Women in Engineering (WMWE)* Sep 2026 – Present
- Leading a mentoring team of **5+** through meetings, discussions, and events that advance women in engineering at UT Dallas.
- Partnering with Mentoring Team Coordinators to cultivate an inclusive environment and meaningful mentor–mentee connections.

Industry Committee Officer | *Society of Women Engineers (SWE)* Aug 2026 – Present
- Coordinating technical workshops and info sessions with corporate technology representatives for **50+** engineering students.
- Spearheading sponsorship outreach and event logistics that connect students to software engineering career pathways.

Founder & President | *Women in STEM Club* Jun 2025 – May 2026
- Founded a STEM initiative, teaching foundational coding through hands-on workshops for **40+** middle and high school students.
- Hosted virtual STEM sessions and student-focused initiatives that sustained interest in technology and computer science.

Honors & Awards

Winner, Best Business Innovation – EmpowerHer Roseville Hackathon (BuildHer Finance) Jul 2025
Winner, Best of Honors – CVHS Hackathon (Voice2Learn) Jul 2025
Finalist – Unity Hackathon (LegalSHEield) Jul 2025
Finalist – CADpossible Hackathon (SenseConnect) Jul 2025